Roshan Syed

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SUMMARY

Data Science and Machine Learning professional with production experience building end-to-end ML scoring pipelines, automating ETL workflows on AWS, training generative diffusion models, and deploying cloud-native prediction systems. Skilled in feature engineering, statistical modeling, deep learning fine-tuning, cloud orchestration, and scalable data infrastructure. Pursuing M.S. in Data Science at the University of Maryland, College Park.

TECHNICAL SKILLS

Programming Languages: Python, R, SQL, HTML

Data Analysis & ML: Scikit-learn, LightGBM, XGBoost, Pandas, NumPy, Feature Engineering, Ensemble Methods, K-Means Clustering, Regression, Classification, Time Series Forecasting

Deep Learning & GenAI: PyTorch, TensorFlow, Keras, CNNs, LSTMs, Transformers (BERT), Transfer Learning, LoRA Fine-Tuning, Stable Diffusion, ControlNet, LangChain, RAG Pipelines

Computer Vision & NLP: OpenCV, YOLO v3, SSD, Image Super-Resolution, Object Detection, Sentiment Analysis, Real-ESRGAN, U2-Net

Data Engineering: ETL/ELT Pipelines, Parquet, MySQL, PostgreSQL, Data Cleaning, EDA, Statistical Analysis, Hypothesis Testing

Cloud & MLOps: AWS (S3, RDS, Lambda, ECS Fargate, Step Functions, EventBridge, SNS), Docker, Git, CI/CD, Gradio, Hugging Face

Data Visualization: Power BI, Tableau, Matplotlib, Seaborn

Mathematics & Statistics: Probability & Statistics, Linear Algebra, Hypothesis Testing, A/B Testing, Statistical Inference

WORK EXPERIENCE

Machine Learning Intern | *Fyenn Labs Services* Feb 2024 – May 2024

- Architected and deployed **Ultra-Focus**, a low-latency image super-resolution inference pipeline leveraging deep convolutional upsampling networks, achieving **2× resolution enhancement on live surveillance feeds** while eliminating hardware upgrade dependencies, reducing per-camera infrastructure cost by ~40%.
- Engineered **Ultra-Focus 2.0**, optimizing batch and streaming inference pipelines for IoT camera ingestion — implemented frame-level preprocessing, model serving optimization, and asynchronous I/O to **reduce end-to-end frame processing latency by 35%** across distributed deployments.
- Built unsupervised market segmentation pipeline using **K-Means clustering with silhouette-score-based k-selection** on consumer behavioral features, identifying 4 actionable buyer segments that informed EV adoption targeting under India's subsidy scheme.
- Delivered quantitative demand-analysis reports across **3 vertical sectors** (security, enterprise, education), synthesizing competitive landscape data and total addressable market sizing for executive stakeholders.

PROJECTS

Siddhantha T20: Stock Intelligence Platform | *LightGBM, AWS (S3, RDS, Lambda, ECS, Step Functions), MySQL* 2025 – Present

- Architected **production-grade ML scoring pipeline processing 3,000+ tickers daily** across Russell 2000, S&P 500, and S&P 600 universes, training per-universe LightGBM binary classifiers with time-series-aware train/validation splits targeting 8–20% forward-return thresholds.
- Engineered **100+ predictive features** across multi-window technical indicators (momentum, volatility, OHLCV ratios, market-relative beta) and cross-sectional fundamental factors (value, growth, quality, leverage) with z-score normalization over **11+ years of historical data** serialized in Parquet on S3.
- Deployed fully automated cloud-native pipeline: containerized **ECS Fargate tasks orchestrated via Step Functions** with EventBridge-scheduled daily/weekly cadences, SNS per-task failure alerting, RDS (MySQL) prediction persistence, and Lambda-served dashboard with OAuth-gated access control.

ClearShot: AI-Powered Product Photo Enhancement | *Python, PyTorch, Stable Diffusion 1.5, ControlNet, LoRA, Real-ESRGAN, Docker, Gradio* Spring 2026

- Built end-to-end **5-stage generative restoration pipeline** (U2-Net background segmentation, Canny structural conditioning, ControlNet-guided Stable Diffusion 1.5 image-to-image enhancement, studio compositing, Real-ESRGAN 2× super-resolution) transforming amateur smartphone product photos into catalog-quality output on a curated **24,131-image Amazon Berkeley Objects subset**; deployed publicly on Hugging Face Spaces with Docker containerization.
- Authored LoRA fine-tuning script **(rank=16, α=32, targeting UNet cross-attention to_q/to_k/to_v/to_out.0 projections)** integrated with accelerate.Accelerator for mixed-precision training and gradient accumulation; trained adapter for 5,000 optimizer steps on Colab T4, reducing MSE loss from 0.0802 to 0.0701 with monotonic smoothed convergence.
- Diagnosed and resolved silent LoRA-loading compatibility bug by implementing a custom PeftModel.from_pretrained() helper after the default diffusers loader failed to parse PEFT-format adapter keys; full pipeline achieves **best-in-class LPIPS 0.322 and FID 88.66** across 5 evaluated baselines on a 457-image stratified test set.
- Implemented DiffusionEnhancer wrapper with **UniPCMultistepScheduler**, xformers memory-efficient attention with attention-slicing fallback, and fp16/fp32 hardware abstraction for portable CPU/GPU inference.

AI-Driven Amazon Product Suggestions through Object Detection | *Python, OpenCV, YOLO v3, TensorFlow, Pandas* Fall 2024

- Architected **real-time visual product recommendation system** combining YOLO v3 object detection with a curated Amazon catalog of **10,000+ product entries across 7 macro-categories**, delivering image-based product suggestions with bounding-box transparency and direct purchase URLs.
- Engineered YOLO inference pipeline using **Darknet-53 backbone with three-scale detection**: blob preprocessing at 416×416, confidence-threshold filtering (0.5), and Non-Max Suppression for bounding-box refinement; trained for 10 epochs achieving **86.93% final accuracy, 0.94 recall**, and peak 94.75% accuracy at epoch 6.
- Built **keyword-matching engine** mapping detected COCO class labels to Amazon product entries via case-insensitive string matching, enabling automated linkage of visual inputs to product metadata (name, category, price, URL).
- Performed dataset preprocessing across 10K product entries: missing-value imputation, price-range averaging (e.g., "$20–$40" → midpoint), and category standardization; produced EDA artifacts (histograms, boxplots, category pie charts) to inform price-segmentation strategy.

Emojify: Real-Time Facial Emotion Recognition | *Python, PyTorch, OpenCV, CNN* 2024

- Trained custom **5-layer convolutional neural network from scratch on FER-2013** (35,887 grayscale facial images across 7 emotion classes), achieving 65%+ test accuracy through architecture tuning and data augmentation.
- Built **end-to-end real-time classification pipeline** integrating webcam capture, Haar Cascade face detection, CNN inference, and emoji response rendering; achieved **sub-150ms latency on CPU** for live deployment.

EDUCATION

Master of Science in Data Science 2024 – 2026
University of Maryland, College Park GPA: 3.9 / 4.0

Bachelor of Engineering – Artificial Intelligence & Machine Learning 2020 – 2024
New Horizon College of Engineering CGPA: 8.46 / 10.0